

Web Application Development

JavaScript



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN



-  Understand the significance of JavaScript in a website.
-  Know the principles and syntax of JavaScript programming.
-  Know how to use JavaScript to program and handle events in a website.



Introduction to JavaScript

Embedding JavaScript in a Web Page

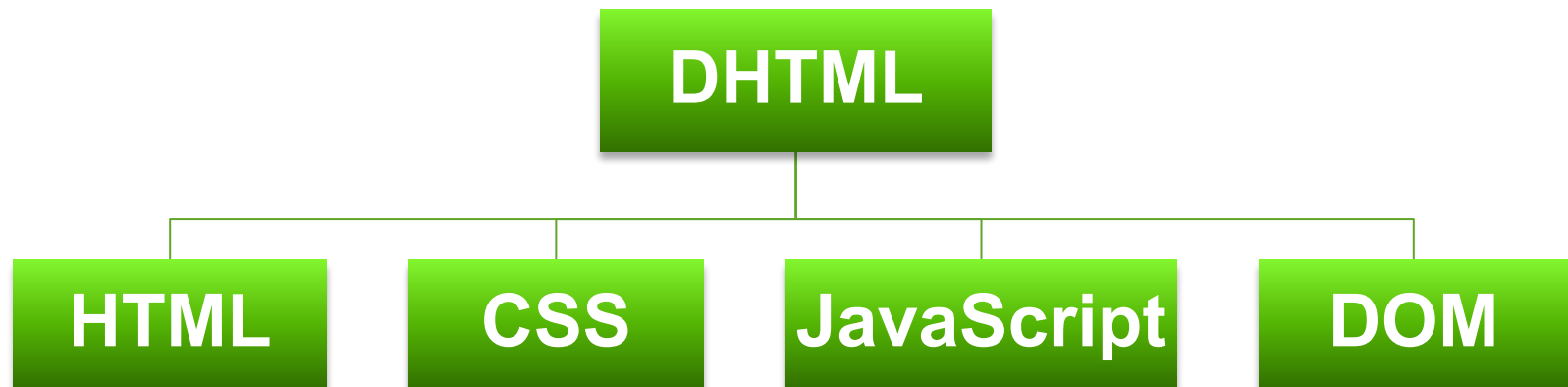
Data Types & Syntax in JavaScript

Event Handling

HTML DOM with JavaScript



- Allows the website to interact and change according to user actions.
- $\text{DHTML} = \text{HTML} + \text{CSS} + \text{JavaScript}$





☐ HTML

- ☐ Defines the content of the web page through semantic tags
 - heading, paragraph, list, etc.

☐ CSS

- ☐ Defines the rules or styles to display the HTML document:
 - Fonts
 - Background (color, image, etc.)
 - Position and size

☐ JavaScript

- ☐ Defines actions:
 - Interaction through user actions, event handling, etc.



- It is a client-side scripting language that runs on the user's browser (client).
- It shares processing in web applications, reducing unnecessary processing on the server.
- It helps create effects and interactions for the website.



Client-Side Script:

- ☐ Scripts executed on the client-side (browser):
Perform interactions with users (create animated menus, etc.), validate input data, etc

Server-Side Script:

- ☐ Scripts processed on the server-side aim to create web pages capable of generating dynamic content. Some main processes include:
connecting to databases, accessing the file system on the server, generating HTML content to return to the user, etc



Define the script directly within the HTML page.

```
<script>  
<!--  
// JavaScript  
-->  
</script>
```

Embed using a script installed from another .js file

```
<script src="xxx.js"></script>
```




- Place it between the `<head>` and `</head>` tags: the script will execute as soon as the webpage is opened.
- Place it between the `<body>` and `</body>` tags: the script in the body section will execute while the webpage is loading (after executing the scripts in the `<head>` section).
- There is no limit to the number of client-side script sections you can embed in the page.



Variable naming conventions

- ☐ Begin with a letter or the symbols `_`, `$`
- ☐ `A..Z`, `a..z`, `0..9`, `_`: **case-sensitive**



Variable declaration

- ☐ Use the **var** keyword to limit the scope of the variable within a function.
- ☐ Use the **let** and **const** keywords to limit the scope of variables within a code block.
- ☐ There is no need to declare a variable before using it; the variable will actually exist when it is first used and will have a global scope.



- ☐ The scope of a variable is limited within a function (**var**) or a block (**const**, **let**).
- ☐ All variables declared outside of a function have global scope.
- ☐ All global variables are children of the window object.
- ☐ If a variable is assigned without being declared, it will have global scope.

```
function a(){  
  if (1==1){  
    var x = 10;  
  }  
  console.log(x);  
}
```

```
function a() {  
  x = 10;  
}  
a();  
console.log(x);
```



Kiểu dữ liệu	Ví dụ	Mô tả
object	<code>var listBooks = new Array(10) ;</code>	trước khi sử dụng, phải cấp phát bằng từ khóa new
string	<code>"The cow jumped over the moon."</code> <code>"40"</code>	Chứa được chuỗi unicode Chuỗi rỗng <code>"</code>
number	0.066218 12	Theo chuẩn IEEE 754
boolean	true / false	
undefined	<code>var myVariable ;</code>	<code>myVariable = undefined</code>
null	<code>connection.Close();</code>	<code>connection = null</code>



- A variable automatically changes its data type when the value it stores changes

```
var x = 10; // x is a number
```

```
x = "hello world !"; // x is now a string
```

- It is possible to concatenate two variables with different data types.

```
var x;
```

```
x = "12" + 34.5; // x = "1234.5"
```

- Functions parseInt(...), parseFloat(...):
Convert data types from string to number.



- onclick
- onfocus
- onchange
- onblur
- onmouseover
- onmouseout
- onmousedown
- onmouseup
- onload
- onsubmit
- onresize
-



Syntax

```
<tag eventhandler = "JavaScript  
Code">
```

```
<body>
```

```
<input type="button" name="Button1"  
value="OpenSesame!"
```

```
onclick="window.open('mydoc.html');">
```

```
</body>
```



```
<head>
  <script language="Javascript">
    function GreetingMessage ()
    {
      window.alert("Welcome to my world");
    }
  </script>
</head>

<body onload="GreetingMessage ()">
</body>
```




Assign a callback function for an object event

`object.eventhandler = function_name;`

```
<head>
  <script language="Javascript">
    function GreetingMessage()
    {
      window.alert("Welcome to my world");
    }

    window.onload = GreetingMessage ();
  </script>
</head>

<body>
</body>
```



```
<!DOCTYPE html>
<html>
<head>
  <title>Square calculation</title>
</head>
<body>
<form name="frm">
  X =
  <input type="text" name="expr" size="15">
  <br />
  <br />
  <input type="button" value="Calculate" onclick="compute(this.form)" />
  <br />
  X * X = <span id="result"></span>
</form>
<script>
  function compute(frm) {
    var x = frm.expr.value;
    document.getElementById("result").innerHTML = x * x;
  }
</script>
</body>
</html>
```



- DOM = Document Object Model
- A set of standard HTML objects used to access and modify HTML elements on a webpage (to change the document content of the page).
- Some DOM objects include: window, document, history, link, form, frame, location, event, etc.



-  It is an instance of the browser window object.
-  It exists when an HTML document is opened.
-  It is used to access information about the objects in the browser window (such as the browser name, browser version, title bar, status bar, etc.)



Properties

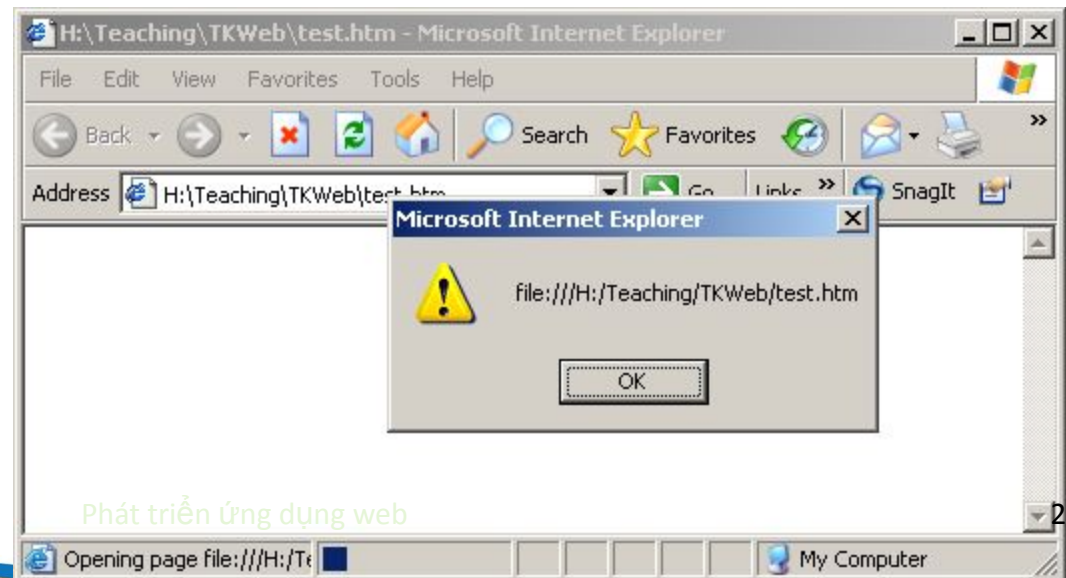
- ☐ document
- ☐ event
- ☐ history
- ☐ location
- ☐ name
- ☐ navigator
- ☐ screen
- ☐ status

- Methods

- ☐ alert
- ☐ confirm
- ☐ prompt
- ☐ blur
- ☐ close
- ☐ focus
- ☐ open



```
<html>
  <body>
    <script type="text/javascript">
      var curURL = window.location;
      window.alert(curURL) ;
    </script>
  </body>
</html>
```





- Represents the content of the HTML page currently displayed in the browser.
- Used to retrieve information about the document, HTML elements, and handle events.





- Properties

- aLinkColor
- bgColor
- **body**
- fgColor
- linkColor
- title
- **URL**
- vlinkColor
- forms[]
- images[]
- **childNodes[]**

- **documentElement**
- cookie
-

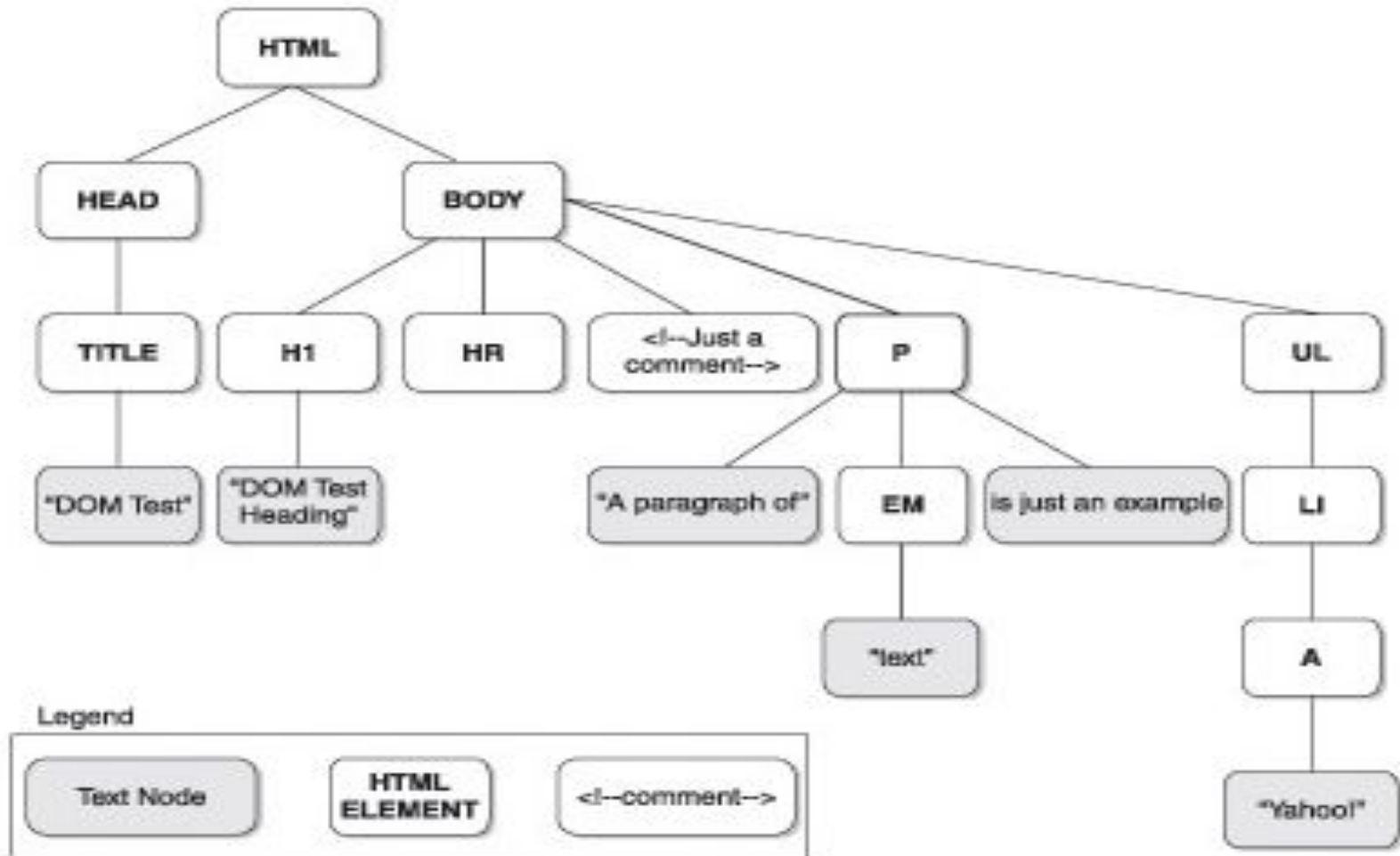
- Methods

- close
- open
- **createTextNode(" text ")**
- **createElement("HTMLtag")**
- **getElementById("id")**
- ...



Represents the content of the document in a tree structure

```
<html>
<head>
  <title>DOM Test</title>
</head>
<body>
  <h1>
    DOM Test Heading</h1>
  <hr />
  <!-- Just a comment -->
  <p id="p1">
    A paragraph of <em>text</em> is just an example</p>
  <ul>
    <li><a href="http://www.yahoo.com">Yahoo! </a></li>
  </ul>
</body>
</html>
```





getElementById (id1)

```
//<p id="id1" >  
//some text  
//</p>
```

```
var node = document.getElementById("id1");  
var nodeName = node.nodeName; // p  
var nodeType = node.nodeType; // 1  
var nodeValue = node.nodeValue; // null  
var text = node.innerText ; // some text
```



innerHTML

```
//<p id="para1" >  
//some text  
//</p>  
var theElement = document.getElementById("para1");  
theElement.innerHTML = "Some <b> new </b> text";
```

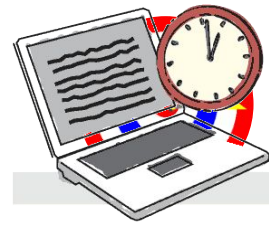
```
// Kết quả :  
// <p id="para1" >  
// Some <b> new <b/> text  
// </p>
```



innerText

Ví dụ:

```
var theElement = document.getElementById("para1");  
theElement.innerText = "Some <b> new </b> text";  
// Kết quả hiển thị trên trình duyệt  
// bên trong thẻ p: "Some <b> new </b> text"
```



- Use HTML, CSS, and JavaScript to create a webpage as follows:

Bé tập tính

Số thứ nhất		☐ Cộng	☐ Trừ
Số thứ hai		☐ Nhân	☒ Chia
Kết quả		Tính	
Thông báo	Giá trị nhập ở ô <i>Số thứ nhất</i> không phải là số		



- Use a CSS framework to create the interface, aligning elements in columns, and allowing only one of the four operations to be selected (2 points).
- Write JavaScript code so that when the calculate button is clicked, the result of the selected operation will be filled in the result box (4 points).
- The notification area will display error messages when:
 - Moving away from the input box for the first number or the second number with data that is not a valid decimal number.
 - Clicking the Calculate button without selecting an operation.
 - Not entering two valid numbers to perform the calculation (3 points).
- Upload to a public host (1 point).



- ☐ http://www.w3schools.com/js/js_object_prototypes.asp
- ☐ <https://developer.mozilla.org/en-US/docs/Web/JavaScript/Closures>
- ☐ http://www.w3schools.com/js/js_function_closures.asp
- ☐ http://localjs.org/docs/en/tutorial_js4cpp.php
- ☐ http://en.wikipedia.org/wiki/Comparison_of_JavaScript_frameworks
- ☐ <http://www.netmagazine.com/features/essential-javascript-top-five-mvc-frameworks>